

Sarah Connor

AI Security Engineer · Adversarial ML & Red Teaming
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SUMMARY

AI security engineer with six years spanning applied machine learning, security engineering, and detection engineering. Builds defenses against adversarial and prompt-injection attacks on production ML systems, and leads red-team exercises against internal LLM deployments.

EXPERIENCE

Senior AI Security Engineer 2023 – Present
Cyberdyne Systems Austin, USA

- Leading red-team engagements against internal LLM-powered products before launch.
- Built an automated prompt-injection fuzzing harness that surfaced 40+ high-severity jailbreaks across 6 production LLM features prior to release.
 - Led adversarial robustness reviews for a fraud-detection model serving 2M inference requests/day, reducing evasion success rate from 18% to under 2%.
 - Designed and shipped a model-extraction detection pipeline (query-pattern anomaly scoring) now running across all customer-facing inference endpoints.
- Python · PyTorch · Burp Suite · Kubernetes · OPA/Gatekeeper

Security Engineer, Applied ML 2021 – 2023
Skynet Analytics Remote

- Owned the security review process for the ML platform team’s model deployment pipeline.
- Authored the org’s first threat model for LLM-integrated services, adopted as the standard review template company-wide.
 - Cut mean time to patch critical model-serving CVEs from 21 days to 4 by embedding security gating directly into the MLOps CI/CD pipeline.
 - Ran quarterly red-team exercises against internal chatbots, discovering data-exfiltration paths later closed via output filtering and scoped tool permissions.
- Terraform · AWS · MLflow · Semgrep · Snyk

Machine Learning Engineer 2019 – 2021
Tyrell ML Ljubljana, Slovenia

- Built and shipped machine learning models powering fraud detection and demand forecasting for a fast-growing fintech product.
- Built and shipped fraud-detection models into production, reducing false-positive rates by 25% across the company’s core payments pipeline.
 - Designed the team’s first automated retraining pipeline, cutting model refresh time from weeks to days.
- TensorFlow · scikit-learn · Docker

SKILLS

Security	Threat modeling, Burp Suite, Nmap, Metasploit
ML/AI	PyTorch, vLLM, adversarial robustness, prompt-injection defense
Infra	Kubernetes, Terraform, AWS, CI/CD pipeline hardening
Practice	Incident response, vulnerability research, red-team exercise design

>	EDUCATION	
	M.S. Computer Security	2023
	Stanford University	Stanford, USA
	B.S. Computer Science	2019
	University of Ljubljana	Ljubljana, Slovenia
>	PROJECTS	
	promptfirewall	2023
	github.com/sconnor/promptfirewall	
>	CERTIFICATIONS, AWARDS & PUBLICATIONS	
	OSCP – Offensive Security Certified Professional	2019
	Offensive Security	
	AI Village CTF – 1st Place	2024
	DEF CON	
	Detecting Prompt Injection at Scale	2024
	arxiv.org/abs/2403.09217	